# Weikang (Zachary) Meng

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## EDUCATION

**Harbin Institute of Technology, Shenzhen** **Sep. 2023 – Present (Expected Dec. 2027)**

Ph.D. in Computer Science and Technology, School of Computer Science and Technology

- Advisor: Prof. Zheng Zhang
- Research Interests: **Linear Attention, Efficient Foundation Models**

**Harbin Institute of Technology** **Sep. 2019 – Jul. 2023**

B.S. in Information and Computational Science, School of Mathematics

- GPA: 85.98/100; Recommended for Direct Ph.D. Admission

## SELECTED PUBLICATIONS & RESEARCH EXPERIENCE

**PolaFormer++: Polarity-aware Channel-Spiky Linear Attention for Vision Transformers (TPAMI 2026, CCF-A TOP, IF: 20.8) [Link]**

Authors: ***Weikang Meng***, Liangyu Huo, Yadan Luo, Yaowei Wang, Yingjian Li, Zheng Zhang, Heng Tao Shen

- Extended PolaFormer with channel-wise spikiness modeling to improve attention sharpness and feature discrimination, which demonstrated strong transferability across multiple vision tasks.

**Norm Direction: Restoring the Missing Query Norm in Vision Linear Attention (ICML 2026, CCF-A) [Link]**

Authors: ***Weikang Meng***, Yadan Luo, Liangyu Huo, Yaowei Wang, Xin Li, Zheng Zhang

- Revealed the missing-query-norm in linear attention and derived Positive Sequence Entropy theory.
- Proposed norm-aware kernel functions and geometric similarity functions, forming a novel linear attention.

**PolaFormer: Polarity-aware Linear Attention for Vision Transformers (ICLR 2025, CCF-A, 100+citations)[Link]**

Authors: ***Weikang Meng***, Yadan Luo, Xin Li, Dongmei Jiang, Zheng Zhang

- Proposed polarity decomposition to explicitly model positive and negative Query-Key interactions.
- Established theoretical conditions for sharper attention distributions, achieving SOTA performance.

**STILL: Selecting Tokens for Intra-Layer Hybrid Attention to Linearize LLMs (NeurIPS 2026, CCF-A) [Link]**

Authors: ***Weikang Meng***, Liangyu Huo, Yadan Luo, Jiawen Guan, Jingyi Zhang, Yingjian Li, Zheng Zhang

- Proposed intra-layer hybrid attention for linearizing LLM, with softmax attention and linear attention.
- Developed Self-Saliency Score for Token selection and Norm-Preserved Feature Map for linear branches, which improving long-context efficiency and restoring model capability.

**Cluster-Wise Spatio-Temporal Masking for Efficient Video-Language Pretraining (CVPR 2026, CCF-A) [Link]**

Authors: W Zhuang, Y Huang, ***Weikang Meng,*** X Li, M Liu, X Hong, Y Wang, W Zuo

- Cluster-wise spatio-temporal masking for preserving holistic video content and temporal consistency.
- Video-text relevance reconstruction for aligning high-level multimodal semantics and improving video-language representations.

**TOM-Pruning: Target-aware Output Manifold for LLM Pruning (NeurIPS 2026, CCF-A)**

Authors: Junchen Hao, ***Weikang Meng,*** Yingjian Li, Zheng Zhang

- Reframes post-training LLM pruning from a novel target-aware output-manifold perspective.
- Introduces a Target-Aware Local Riemannian Metric and a Competitive-Hubness Channel Affinity Mechanism.

**Ascend Efficient Transformer Architecture Optimization (Huawei Joint Research Project, 2023, 1.2M funding)**

- Designed an Ascend-friendly linear-complexity attention to replace softmax attention, cutting per-step training time by 20% and memory by 15% on BERT/GPT2/LLaMA-scale models using mindspeed-llm.
- Honored with the HUAWEI Spark Award.

*For a complete list of publications, please see my **Google Scholar** profile.*

## PROFESSIONAL SERVICE

**Ascend-TLA: The First Efficient Linear Attention Operator Library for Ascend NPU [Link]**

- The first efficient linear attention operator library for Ascend NPUs; merged into the main repository CANN.
- *CANN Contributor* and *CANN Core Developer.*

**Reviewer/Program Committee: NeurIPS, ICML, ICLR, ECCV, CVPR, AAAI, TIP, TCSVT, PR.**

## HONORS & AWARDS

- Honorable Mention, Mathematical Contest in Modeling (MCM/ICM) (2022)
- Provincial Second Prize, Chinese Mathematics Competition for College Students (2020)
- Provincial First Prize, National High School Mathematics Competition (2018)